

HaptoRoom: Using Vibrotactile Floor Interfaces to Enable Reconfigurable Haptic Interaction onto Any Furniture Surfaces

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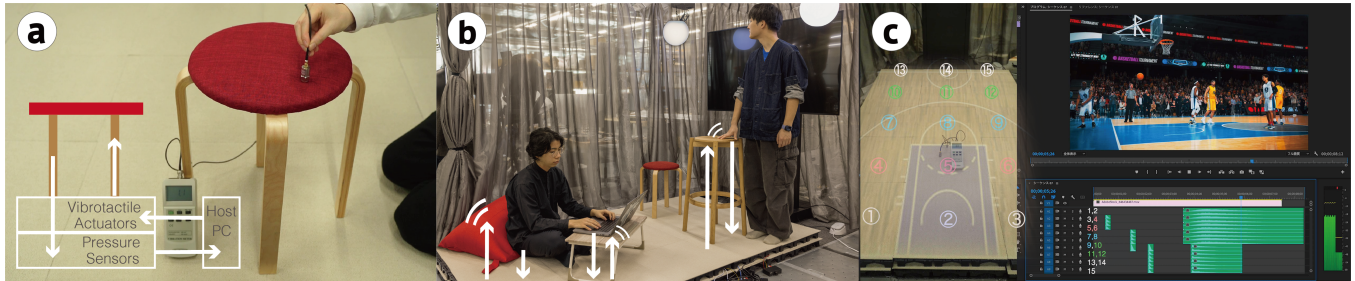


Figure 1: Overview of HaptoRoom system: (a) A chair surface provides vibrotactile feedback to user's touch, without any equipped sensors nor actuators on the chair. Floor interfaces provide pressure sensing and vibrotactile actuation capabilities, which gives haptic interactivity to the furniture. (b) HaptoRoom system can handle multiple haptic-enabled areas at the same time across the room, and (c) the spatial vibrotactile patterns can be designed by off-the-shelf video authoring tools.

ABSTRACT

Vibrotactile interaction is an essential source of HCI and should be on a large scale, such as room scale, whereas in the past many devices have been small. However, there is a practical difficulties for implementing vibrotactile actuators and sensors to every furnitures in the room. To address this problem, our approach, HaptoRoom, archives reconfigurable ubiquitous haptic interaction to existing furniture without additional equipment by installing actuators only in the floor. A user can design detailed spatial distribution of vibrotactile pattern, leveraging with multi-point individual control of actuators array inside floor interfaces.

CCS CONCEPTS

• Human-centered computing → Haptic devices; Ambient intelligence.

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KEYWORDS

haptics design, vibrotactile feedback, smart home, ubiquitous computing, interactive furniture, floor interface

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1 INTRODUCTION

The field of HCI research explores the potential for haptic interaction at the desk, chair, and even room scale. [Ishii et al. 1998] At room scale, it is desirable to provide haptic interaction throughout the entire non-wearable, physical space. However, it is difficult to realize tactile interaction by attaching actuators to all surfaces of a room. This would require specialized furniture, which would be incompatible with existing furniture, a major challenge. Ideally, however, existing furniture should be used as they are, and haptic interactions can be presented on those surfaces.

Our approach to this challenge is to focus on the "floor". In HCI, the floor is a privileged interface because it is in constant contact with people and objects [Visell et al. 2009]. There is research into developing interaction experiences by using the floor as a measurement location [Bränzel et al. 2013]. There are also already proposals to use the floor as a place of measurement and to use all the furniture in a room as touch sensors [Yoshida et al. 2023]. We extend this idea and propose a method to present tactile feedback on furniture surfaces in a room through the coordinated control of a group of vibration actuators installed under the floor.

In this system, vibration feedback can be extended to any piece of furniture in a room without the need for additional equipment. This paper proposes the concept of the system and describes the prototype built. Specific applications of HaptoRoom in entertainment and lifestyle are also proposed.

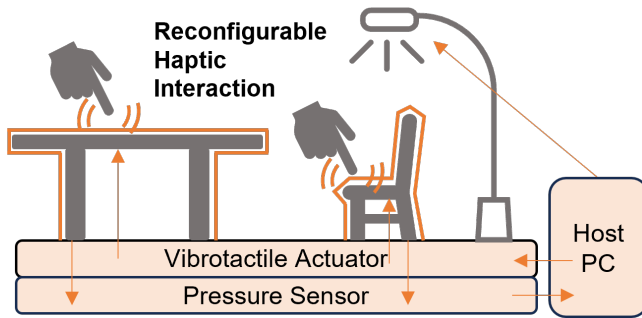


Figure 2: HaptoRoom System Architecture

2 HAPTOROOM SYSTEM

HaptoRoom system introduces haptic interaction onto any furniture surfaces by enabling ordinary furniture, like chairs or tables, to offer vibrotactile interaction without needing sensors or actuators on them. The system architecture is shown in Fig.2. The system supports multiple haptic areas simultaneously and allows for the design of spatial vibrotactile patterns using standard video authoring tools, enhancing user immersion and interaction. Reconfigurable haptic interactions open up the large design space for furniture, people, surface textures and so on.

For our prototype implementation in a 2.5m x 1.5m area, HaptoRoom utilizes floor modules in a 50cm x 50cm square with embedded pressure sensors [Yoshida et al. 2022] and vibrotactile actuators. Controlled by Cycling '74 Max, it enables targeted furniture to provide real-time vibrotactile feedback, with adjustable vibration intensity. This multi-channel control supports numerous haptic feedback zones, enriching the immersive experience.

The system's effectiveness is evaluated by measuring vibrotactile intensity on items like desks and chairs using the off-the-shelf vibrometry, VB-8201HA. As shown in Fig.3, the vibration measurements shows 1.2 m/s² for desk surface, 0.7m/s² for floor surface, and 1.1m/s² for chair surface.

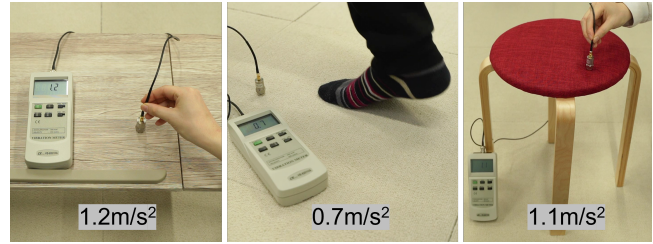


Figure 3: Propagating vibration from floor to the furniture

3 USE SCENARIOS

HaptoRoom is applicable to a wide variety of application areas. Here we show examples in entertainment and lifestyle scenarios.

Especially for sports games as an entertainment application, HaptoRoom can enhance the viewer's immersive experience. Multiple people can experience together while watching the game in individually selected locations, such as chairs or cushions. Additionally, when the spatial orientation of the room mirrors the layout of the sports court, viewers can physically sense the position of the players and ball relative to their position.

For the lifestyle scenarios, one of the key advantages of vibration feedback is that it allows for a personalized approach to individual users. Sound alarms may wake not only the intended individual, but also others in the surrounding area. With HaptoRoom, however, vibration feedback can be used as a personal alarm. HaptoRoom interfaces furniture by integrating vibration feedback into the furniture. For example, a chair can be used as a trigger to control the lighting in the room. A user will find themselves acclimating in no time, as the haptic cues guide their interactions, making the space uniquely responsive to their touch and presence.

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REFERENCES

- Alan Bränzel, Christian Holz, Daniel Hoffmann, Dominik Schmidt, Marius Knaust, Patrick Lühne, René Meusel, Stephan Richter, and Patrick Baudisch. 2013. *GravitySpace: Tracking users and their poses in a smart room using a pressure-sensing floor*. 734 pages. <https://doi.org/10.1145/2470654.2470757> journalAbbreviation: Conference on Human Factors in Computing Systems - Proceedings.
- Hiroshi Ishii, Craig Wisneski, Scott Brave, Andrew Dahley, Matt Gorbett, Brygg Ullmer, and Paul Yarin. 1998. *ambientROOM: integrating ambient media with architectural space*. In *CHI 98 Conference Summary on Human Factors in Computing Systems*. ACM, Los Angeles California USA, 173–174. <https://doi.org/10.1145/286498.286652>
- Yon Visell, Alvin Law, and Jeremy R. Cooperstock. 2009. *Touch Is Everywhere: Floor Surfaces as Ambient Haptic Interfaces*. *IEEE Transactions on Haptics* 2, 3 (July 2009), 148–159. <https://doi.org/10.1109/TOH.2009.31>
- Takatoshi Yoshida, Narin Okazaki, Ken Takaki, Masaharu Hirose, Shingo Kitagawa, and Masahiko Inami. 2022. *Flexel: A Modular Floor Interface for Room-Scale Tactile Sensing*. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology (UIST '22)*. Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3526113.3545699>
- Takatoshi Yoshida, Takafumi Watanabe, Hirosuke Asahi, Kashun Rin, Haruki Kozuka, Masaharu Hirose, and Masahiko Inami. 2023. *ForceField: Visualizing Intermaterial Interaction through Floor and Depth Sensing*. In *SIGGRAPH Asia 2023 Emerging Technologies (SA '23)*. Association for Computing Machinery, New York, NY, USA, 1–2. <https://doi.org/10.1145/3610541.3614587>